MIL-S-8932/2
12 August 1980
SUPERSEDING
MIL-S-8934 .
28 January 1965

MILITARY SPECIFICATION SHEET

SWITCHES, PRESSURE, AIRCRAFT, DIFFERENTIAL,
TYPES I AND II, CLASS 2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the switch assembly described herein shall consist of this document and the issue in effect of specification MIL-S-8932.

1. SCOPE

1.1 Scope. This specification covers the detail requirements of aircraft differential pressure switches for use in ac and dc applications.

1.2 Classification. Pressure switches covered by this specification shall be of the following types and class as specified.

Type - Types I and II switches shall be as specified in MIL-S-8932 (see 6.2).

Class - Class 2 switches shall be the aircraft differential pressure switches which respond to the difference in pressure between two applied pressures.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to : Commanding Officer, Naval Air Engineering Center, ESSD, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1650

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SPECIFICATIONS

Military

MIL-S-8932	Switches, Pressure, Aircraft, General Specification for

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with the general specification MIL-S-8932 and as specified herein. Each switch supplied under this specification shall incorporate a manufacturer's part number and shall have passed the first article tests as specified in MIL-S-8932.

3.1.1 Pressure port designations. The capital letter "P" shall be permanently and legibly stamped on the switch case directly adjacent to the switches pressure port.

3.2 Design.

3.2.1 Case. Dimensions shall conform to the applicable MS.

3.2.2 Number of contacts. Number of contacts shall be as illustrated on the applicable figure 1 or 2.

3.2.3 Internal wiring. The switch contacts shall be wired to the connector terminals as illustrated on the applicable figure 1 or 2.

3.2.4 Electrical connector. The electrical connector used shall be as illustrated on the applicable figure 1 or 2.

3.2.5 Operating range. The range throughout which the switch must be adjustable as specified on the applicable MS.

3.3 Current ratings and types of load.

3.3.1 Alternating current (at). The maximum value of ac current that the switch is required to carry shall be as specified on the applicable MS. The load shall be resistive.

3.3.2 Direct current (de). The maximum value of direct current (dc) that the switch is required to carry shall be as specified on the applicable MS.

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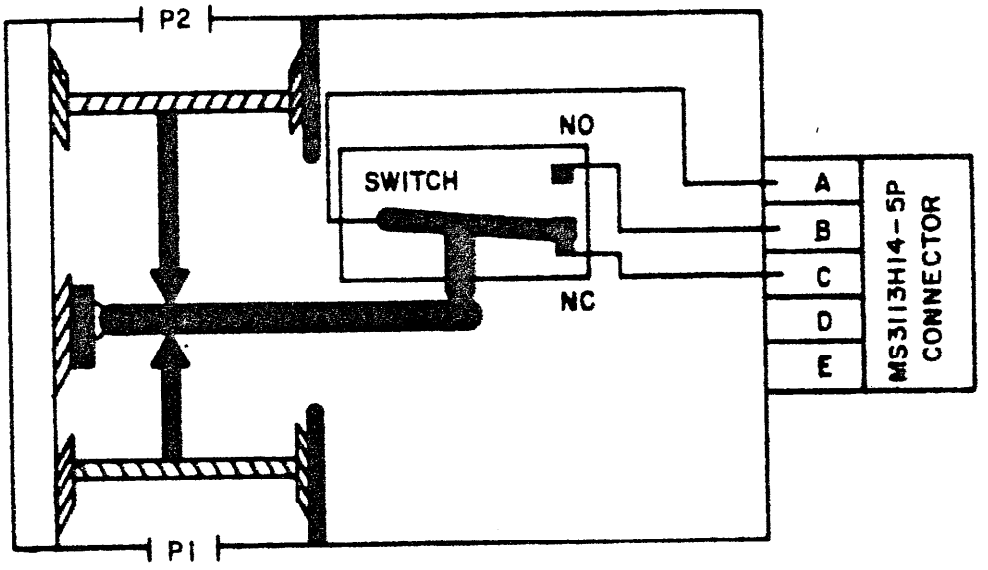


Figure 1. Differential pressure switch schematic, pressure port open.

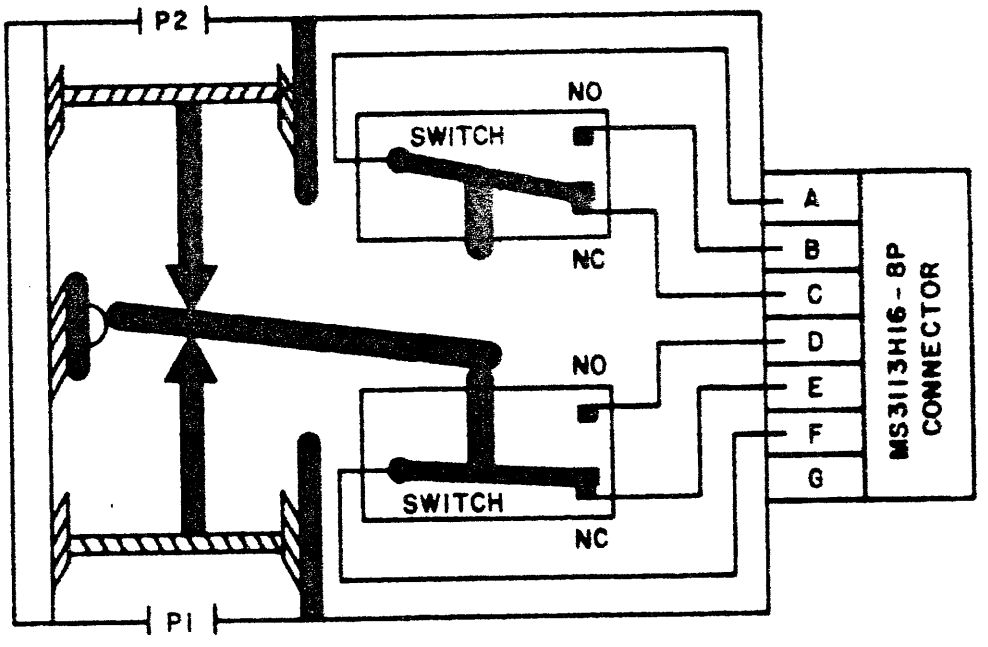


Figure 2. Differential pressure switch schematic, pressure port open.

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3.3.3 Identification of product. A nameplate, permanently and legibly marked with the following information, shall be securely attached to the switch case. The nameplate shall not be attached to, or form an integral part of a cover plate that is removable.

SWITCH, PRESSURE, AIRCRAFT
Type I (Engine mounted) or
Type II (Bulkhead mounted) (as applicable)
Class 2 - Differential
National Stock Number
MS Part Number
Manufacturer's Code and Part Number
Manufacturer's Serial Number

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-8932 and as specified herein.

4.1.1 First article inspection. The first article inspection shall consist of the applicable examinations and tests specified in MIL-S-8932 and as specified herein.

4.1.1.1 First article samples. The first article samples shall consist of three switches of the part number contracted for.

4.1.2 Quality conformance inspection. The quality conformance inspection shall consist of the tests specified in MIL-S-8932 and as specified herein.

4.1.3 Tolerance. Tolerances for the tests and examination-s specified in MIL-S-8932 and herein shall be as specified on the applicable MS.

4.1.4 Reference values. Before subjecting the pressure switch to any test requiring reference values, the reference values shall be obtained by the method specified in MIL-S-8932.

4.2 Proof pressure and drift. The test shall be conducted as specified in MIL-S-8932, except that the pressure ports P1 and P2 shall be connected together while applying the normal system pressure and the proof pressure. This test shall be repeated, except that the P2 port shall be connected to a vacuum (less than 2 inches of mercury absolute) while applying the normal system pressure and the proof pressure to the PI port.

4.3 Sensor balance. The following sequence of the test shall be conducted following the range of adjustment test. The test shall be performed in the order specified in 4.3.1.

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4.3.1 Sensor-matching-differential gage switch. The differential pressure switch, with the P2 port open to the atmospheric pressure, shall have the adjustment set so that the switch actuates when the pressure applied to the P1 port is less than one-fifth of the normal system pressure. The switch shall then be subjected to:

(a) A pressure shall be applied to the P1 port and the pressures at which the switch actuates and deactuates shall be noted. The difference between the two values (operating differential) shall be within the tolerance specified on the applicable MS.

(b) A pressure equal to approximately one-fifth of the normal system pressure shall be applied to the P2 port. The test specified in (a) shall be repeated. The differential pressure at which the switch actuates and deactuates shall not differ from the values noted in (a) by more than the tolerance specified on the applicable MS.

(c) The test specified in (b) shall be repeated, with two-fifths substituted for one-fifth wherever it appears.

(d) The test specified in (b) shall be repeated, with three-fifths substituted for one-fifth wherever it appears.

(e) The test specified in (b) shall be repeated, with four-fifths substituted for one-fifth wherever it appears.

4.3.2 Sensor-matching-differential vacuum switch. Differential pressure switches which operate on pressure less than atmospheric shall have the adjustment set so that the switch will actuate at a differential pressure less than one-fifth of the operating range with a pressure equal to the lowest value on the applicable MS to be applied to the P2 port.

(a) A vacuum with a value less than the highest value specified on the applicable MS shall be applied to the P1 port. The pressure at which the switch actuates and deactuates shall be noted and the operating differential shall be within the tolerance specified on the applicable MS.

(b) The above test shall be repeated with a vacuum applied to the P2 port equal to approximately one-fifth of the highest value specified on the applicable MS. The differential pressure at which actuation occurs shall not differ from the value established in (a) by more than the tolerance specified on the applicable MS.

(c) The test specified in (b) shall be repeated with two-fifths substituted for one-fifth wherever it appears.

(d) The test specified in (b) shall be repeated with three-fifths substituted for one-fifth wherever it appears.

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(e) The test specified in (b) shall be repeated with four-fifths substituted for one-fifth wherever it appears.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be in accordance with MIL-S-8932.

6. NOTES

6.1 Intended use. The aircraft differential pressure switches are intended for use in aircraft, in various ac and dc circuits.

6.2 Ordering data. The instructions and definitions contained in Section 6 of MIL-S-8932 are applicable to, and form a part of this specification.

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DOCUMENT IDENTIFIER (Number) AND TITLE

MIL -S-8932/2 SWITCHES, PRESSURE, AIRCRAFT, DIFERENTIAL , TYPES I & II, CLASS-2

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